

Fractals

Infinite Details from Simple Rules

Speaker 1: Foundations & History

Department of Computer Science and Engineering
Bangladesh University of Engineering and Technology (BUET)

Academic Seminar

A Crisis in 19th-Century Mathematics

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In 1872, **Karl Weierstrass** broke this assumption — on purpose.

He constructed a function that is **continuous everywhere**, yet **differentiable nowhere**: infinitely jagged, at every single scale, with no tangent line anywhere at all.

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For nearly a century, these were considered **beautiful but useless** — curiosities built to stress-test the foundations of mathematics, not to describe anything real.

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- He had a real problem no one had connected to these ideas.
- He had **computers** — something Cantor, Koch, and Julia never had — letting him see these shapes for the first time.
- He united a century of scattered "monsters" under one name: **fractal**, from the Latin *fractus* ("broken").

What are Fractals?

Definition

A **fractal** is a geometric shape containing detailed structure at arbitrarily small scales, usually having a **fractal dimension** strictly exceeding the **topological dimension**.

Key Characteristics:

- **Scale invariance**: Details at every level of magnification
- **Fractal dimension**: More than the *conventional* dimension
- **Infinite complexity**: Generated by simple, recursive rules
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"Clouds are not spheres, mountains are not cones, coastlines are not circles..."

— **Benoit Mandelbrot**

The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

Construction Process

- 1 Start with an equilateral triangle

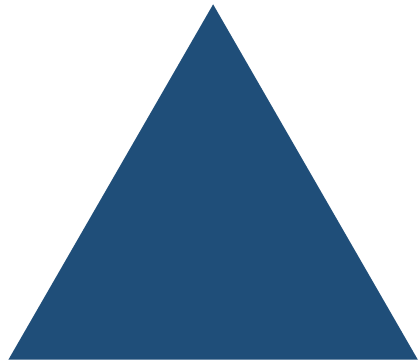


Figure: *

Iteration 0

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Let's construct the Sierpinski Triangle:

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- 1 Start with an equilateral triangle
- 2 Remove the central triangle (triangle connecting midpoints of the sides)

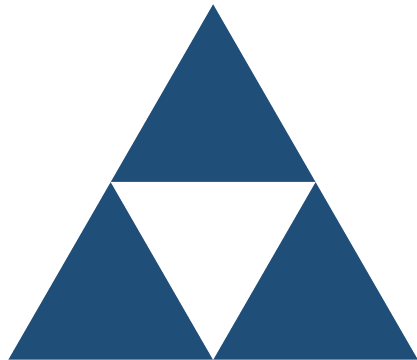


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The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

Construction Process

- 1 Start with an equilateral triangle
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- 3 Repeat for each triangle



Figure: *

Iteration 2

The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

Construction Process

- 1 Start with an equilateral triangle
- 2 Remove the central triangle (triangle connecting midpoints of the sides)
- 3 Repeat for each triangle
- 4 After infinite iterations, we get the Sierpinski Triangle



Figure: *

Iteration 3

The Sierpinski Triangle: Area

Now let's find the area of triangle(s):

Area Calculation

$$A_0 = \frac{1}{2} \cdot \sin(60^\circ) \cdot a^2 = \frac{\sqrt{3}}{4} a^2$$



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$$A_1 = A_0 - \frac{1}{2} \cdot \sin(60^\circ) \cdot \left(\frac{a}{2}\right)^2$$

$$= A_0 - \frac{\sqrt{3}}{16} a^2 = A_0 - \frac{1}{4} A_0$$

$$= \frac{3}{4} A_0$$

We cut out one fourth of the area at each step. So, naturally area is reduced by $\frac{3}{4}$ at each step.



Figure: *

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$$A_0 = \frac{1}{2} \cdot \sin(60^\circ) \cdot a^2 = \frac{\sqrt{3}}{4} a^2$$

$$A_1 = \left(1 - \frac{1}{4}\right) A_0 = \frac{3}{4} A_0$$

$$A_2 = \left(1 - \frac{1}{4}\right) A_1 = \left(\frac{3}{4}\right)^2 A_0$$

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$$A_n = \left(1 - \frac{1}{4}\right)^n A_0 = \left(\frac{3}{4}\right)^n A_0$$

$$\lim_{n \rightarrow \infty} A_n = 0$$

The area of the Sierpinski Triangle is zero even though it is a 2D shape.

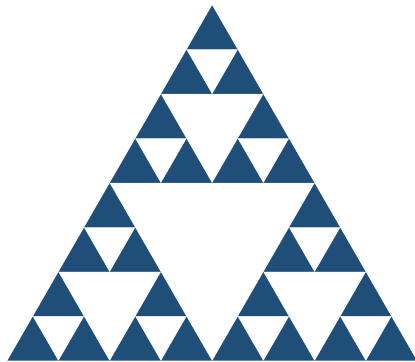


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$$P_0 = 3a$$

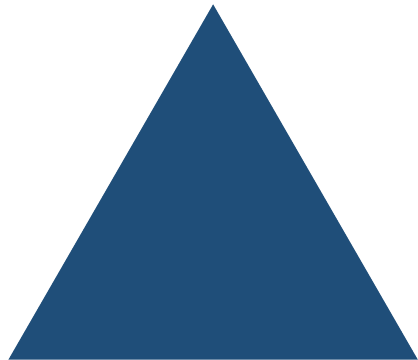


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The Sierpinski Triangle: Perimeter

Now let's find the perimeter:

Perimeter Calculation

$$P_0 = 3a$$

$$P_1 = P_0 + 3 \cdot \frac{a}{2} = \frac{9}{2}a = \frac{3}{2}P_0$$

We add half of the previous perimeter at each step. This is because we add the same number of triangles as the last step, but each triangle is half the size of the previous one. As a result, the perimeter grows by a factor of $\frac{3}{2}$ at each step.



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$$P_n = \frac{3}{2}P_{n-1} = \left(\frac{3}{2}\right)^n P_0$$

$$\lim_{n \rightarrow \infty} P_n = \infty$$

The Sierpinski Triangle has infinite perimeter but zero area.



Figure: *

Iteration 3

Handoff to Speaker 2

We've now seen something strange, and precise:

- A **2D shape** (Sierpinski Triangle) with **zero area**.
- A shape built from a rule so simple a child could apply it by hand.
- No matter how far we zoom in, there's always more detail.

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That number is exactly what Hausdorff was reaching for in 1918 — and it's where Speaker 2 picks up the story.